

## Initial Project Planning Template

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 02 July 2026                    |
| <b>Team ID</b>       |                                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Product Backlog, Sprint Schedule, and Estimation :

Project planning is an essential phase in software development that helps organize the work into manageable tasks and schedules. Agile methodology divides the project into multiple sprints, where each sprint focuses on completing a set of user stories. This approach enables systematic development, continuous testing, and regular progress tracking.

For the **Credit Card Approval Prediction using Machine Learning** project, the work is divided into five sprints covering data preparation, machine learning model development, web application development, testing, and deployment. Each sprint contains user stories that contribute to the successful completion of the project.

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                      | Story Points | Priority | Team Members      | Sprint Start Date | Sprint End Date |
|----------|-------------------------------|-------------------|--------------------------------------------------------------------------------------------------------|--------------|----------|-------------------|-------------------|-----------------|
| Sprint-1 | Dataset Preparation           | USN-1             | As a developer, I want to collect the credit card dataset so that the model can be trained.            | 3            | High     | Anusha            | 20/06/2026        | 20/03/2026      |
| Sprint-1 | Data Preprocessing            | USN-2             | As a developer, I want to clean missing values and preprocess the dataset for model training.          | 5            | High     | Pradeep           | 20/06/2026        | 21/06/2026      |
| Sprint-2 | Feature Engineering           | USN-3             | As a developer, I want to encode categorical features and prepare training data.                       | 3            | High     | Anusha            | 21/06/2026        | 21/06/2026      |
| Sprint-2 | Model Development             | USN-4             | As a developer, I want to train multiple machine learning models and select the best-performing model. | 8            | High     | Sailaja & Pradeep | 22/06/2026        | 24/06/2026      |
| Sprint-3 | Model Evaluation              | USN-5             | As a developer, I want to evaluate model performance using Accuracy, Precision, Recall, and F1-Score.  | 5            | High     | John              | 25/06/2026        | 25/06/2026      |
| Sprint-3 | Model Saving                  | USN-6             | As a developer, I want to save the trained Random Forest model using Joblib for future prediction.     | 2            | Medium   | Sailaja           | 26/06/2026        | 26/06/2026      |
|          |                               | USN-7             | As a user, I want a                                                                                    |              |          |                   |                   |                 |

|          |                            |        |                                                                                                                |   |        |                  |            |            |
|----------|----------------------------|--------|----------------------------------------------------------------------------------------------------------------|---|--------|------------------|------------|------------|
| Sprint-4 | Frontend Development       |        | simple web interface to enter applicant details and receive prediction results.                                | 5 | High   | Pradeep          | 27/06/2026 | 28/06/2026 |
| Sprint-4 | Backend Integration        | USN-8  | As a developer, I want to integrate the Flask application with the trained machine learning model.             | 8 | High   | Anusha & John    | 29/06/2026 | 30/06/2026 |
| Sprint-5 | Testing                    | USN-9  | As a developer, I want to test the application with different applicant inputs to ensure accurate predictions. | 5 | High   | Sailaja          | 30/06/2026 | 30/06/2026 |
| Sprint-5 | Deployment & Documentation | USN-10 | As a developer, I want to prepare documentation and demonstrate the working application.                       | 3 | Medium | All Team Members | 01/07/2026 | 03/07/2026 |